SUBJECT CODE **BOOKLET CODE**

Sr. No. 30583

4 A

2014 (II)
MATHEMATICAL SCIENCES
TEST BOOKLET

E

Time : 3:00 Hours

Maximum Marks: 200

INSTRUCTIONS

1. **You have opted for English as medium of Question Paper.** This Test Booklet contains one hundred and twenty (20 Part'A'+40 Part 'B' +60 Part 'C') Multiple Choice Questions (MCQs). You are required to answer a maximum of 15, 25 and 20 questions from part 'A' 'B' and 'C' respectively. If more than required number of questions are answered, only first 15, 25 and 20 questions in Parts 'A' 'B' and 'C' respectively, will be taken up for evaluation.
2. **OMR** answer sheet has been provided separately. Before you start filling up your particulars, please ensure that the booklet contains requisite number of pages and that these are not torn or mutilated. If it is so, you may request the Invigilator to change the booklet of the same code. Likewise, check the **OMR** answer sheet also. Sheets for rough work have been appended to the test booklet.
3. Write your Roll No., Name and Serial Number of this Test Booklet on the **OMR** Answer sheet in the space provided. Also put your signatures in the space earmarked.
4. **You must darken the appropriate circles with a black ball pen related to Roll Number, Subject Code, Booklet Code and Centre Code on the OMR answer sheet. It is the sole responsibility of the candidate to meticulously follow the instructions given on the OMR Answer Sheet, failing which, the computer shall not be able to decipher the correct details which may ultimately result in loss, including rejection of the OMR answer sheet.**
5. Each question in Part 'A' carries **2** marks, Part 'B' **3** marks and Part 'C' **4.75** marks respectively. There will be negative marking @ 0.5marks in Part 'A' and @ 0.75 marks in Part 'B' for each wrong answer and no negative marking for Part 'C'.
6. Below each question in Part 'A' and 'B', four alternatives or responses are given. Only one of these alternatives is the "correct" option to the question. You have to find, for each question, the correct or the best answer. In Part 'C' each question may have **'ONE'** or **'MORE'** correct options. Credit in a question shall be given only on identification of **'ALL'** the correct options in Part 'C'. No credit shall be allowed in a question if any incorrect option is marked as correct answer.
7. Candidates found copying or resorting to any unfair means are liable to be disqualified from this and future examinations.
8. Candidate should not write anything anywhere except on **OMR** answer sheet or sheets for rough work.
9. Use of calculator is not permitted.
10. **After the test is over, at the perforation point, tear the OMR answer sheet, hand over the original OMR answer sheet to the invigilator and retain the carbonless copy for your record.**
11. Candidates who sit for the entire duration of the exam will only be permitted to carry their Test booklet.

Roll No. 418172

Name Ankur Kanaujiya

OMR Answer Sheet No.....

I have verified all the information filled in by the candidate.

....
Signature of the Invigilator

PART 'A'

1. We define a function $f(N)$ = sum of digits of N, expressed as decimal number. .g. $f(137) = 1 + 3 + 7 = 11$. Evaluate $f(2^7 3^5 5^6)$.
   1. 10
   2. 18
   3. 28
   4. 11

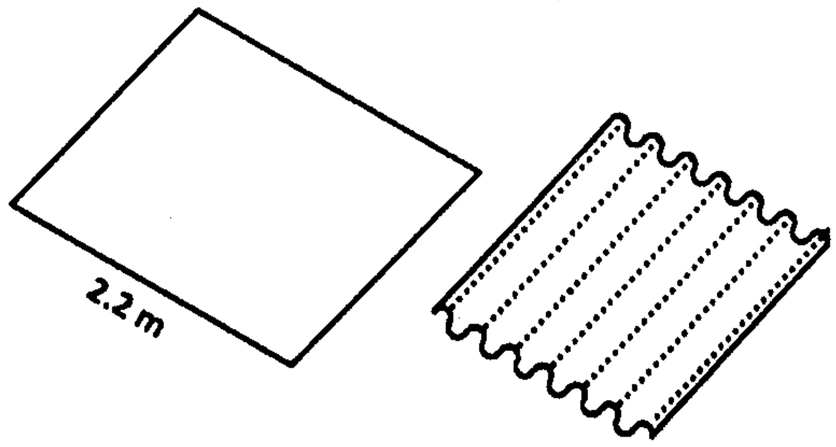
2. Every month the price of a particular commodity falls in this order:
   1024, 640, 400, 250, …
   What is the next value?
   1. 156.25
   2. Approximately 39
   3. 64
   4. 40

3. A cubical piece of wood was filed to make it into the largest possible sphere. What fraction of the original volume was removed?
   1. More than 3/4
   2. 1/2
   3. Slightly less than 1/2
   4. Slightly more than 1/2

4. Three volumes of a Hindi book, identical in shape and size, are next to each other in a shelf, all upright, so that their spines are visible, left to right: I, II and III. A worm starts eating from the outside front cover of volume I, and eats its way horizontally to the outside back cover of volume III. What is the distance travelled by the worm, if each volume is 6 cm thick?
   1. 6 cm
   2. 12 cm
   3. 18 cm
   4. a little more than 18 cm

5. What is the area of the triangle bounded by the lines $y = 2x$, $y = -2x$ and $y = 6$?
   1. 36
   2. 18
   3. 12
   4. 24

6. A 2.2 m wide rectangular steel plate is corrugated as shown in the diagram. Each corrugation is a semi-circle in cross section having a diameter of 7 cm. What will be the width of steel sheet after it is corrugated?



   1. 1.4 m
   2. 1.6 m
   3. 0.7 m
   4. 1.1 m

7. Ajay, Bunty, Chinu and Deb were agent, baker, compounder and designer, but not necessarily in that order. Deb told the baker that Chinu is on his way. Ajay is sitting across the designer and next to the compounder. The designer didn't say anything. What is each person's occupation?
   1. Ajay- compounder; Bunty-designer; Chinu- baker; Deb- agent
   2. Ajay- compounder; Bunty-baker; Chinu- agent; Deb- designer
   3. Ajay- baker; Bunty-agent; Chinu-designer; Deb- compounder
   4. Ajay- baker; Bunty-designer; Chinu-agent; Deb- compounder

8. Two platforms are separated horizontally by distance A and vertically by distance B. They are to be connected by a staircase having identical steps. If the minimum permissible step length is a, and the

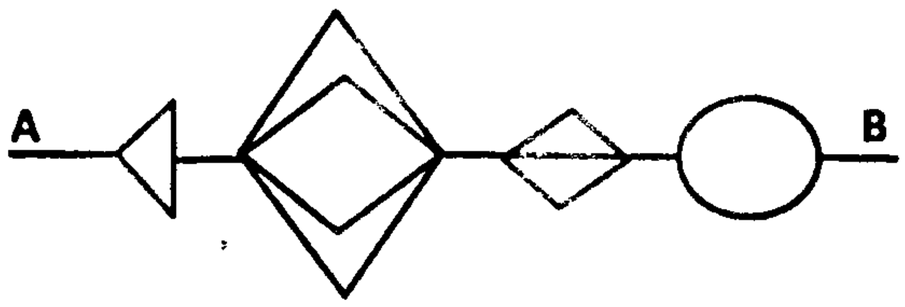
maximum permissible step height is b, the number of steps the staircase can have is

1. $\geq B/b$
2. $\leq A/a$
3. $\geq B/b$ and $\leq A/a$
4. $\leq B/b$ and $\geq A/a$

9. The sum of first n natural numbers with one of them missed is 42. What is the number that was missed?

1. 1
2. 2
3. 3
4. 4

10. A mouse has to go from point A to B without retracing any part of the path, and never moving backwards. What is the total number of distinct paths that the mouse may take to go from A to B?



1. 11
2. 48
3. 72
4. 24

11. A certain day, which is x days before 17th August, is such that 50 days prior to that day, it was $4x$ days since March 30th of the same year. What is x?

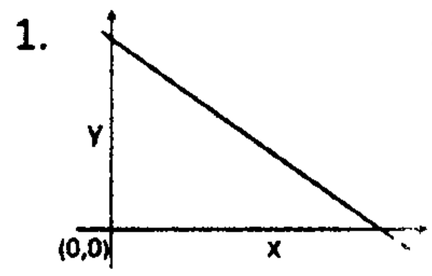
1. 18
2. 30
3. 22
4. 16

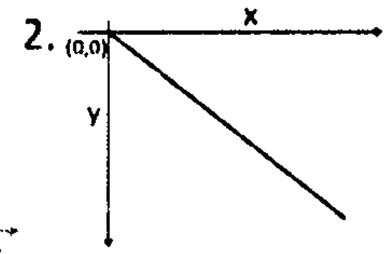
12. What is the next term in the following sequence?

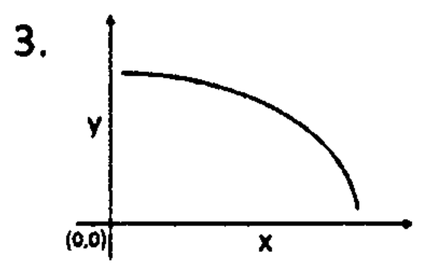
7, 11, 13, 17, 19, 23, 29, ...

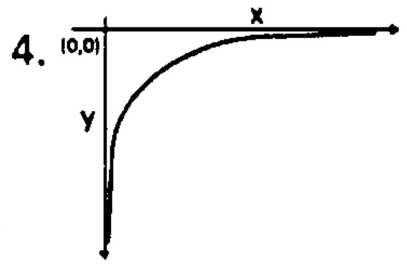
1. 37
2. 35
3. 31
4. 33

13. Which of the following figures best shows that y is inversely proportional to x?

1.


2.


3.


4.


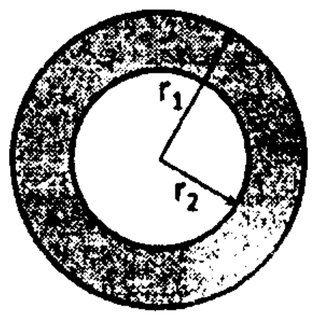
14. What is the maximum number of whole laddoos having diameter of 6 cm that can be packed in a box whose inner dimensions are $24 \times 18 \times 17$ cm^3?

1. 24
2. 30
3. 33
4. 36

15. If N, E and T are distinct positive integers such that $N \times E \times T = 2013$, then which of the following is the maximum possible sum of N, E and T ?

1. 39
2. 2015
3. 675
4. 671

16. The areas of the inner circle and the shaded ring are equal. The radii r_1 and r_2 are related by



1. $r_1 = r_2$
2. $r_1 = r_2\sqrt{2}$
3. $r_1 = r_2\sqrt{3}$
4. $r_1 = 2r_2$

17. The equation $m^2 - 33n + 1 = 0$, where m & n are integers, has

1. no solution
2. exactly one solution
3. exactly two solutions
4. infinitely many solutions

18. Which of the following numbers is a perfect square?

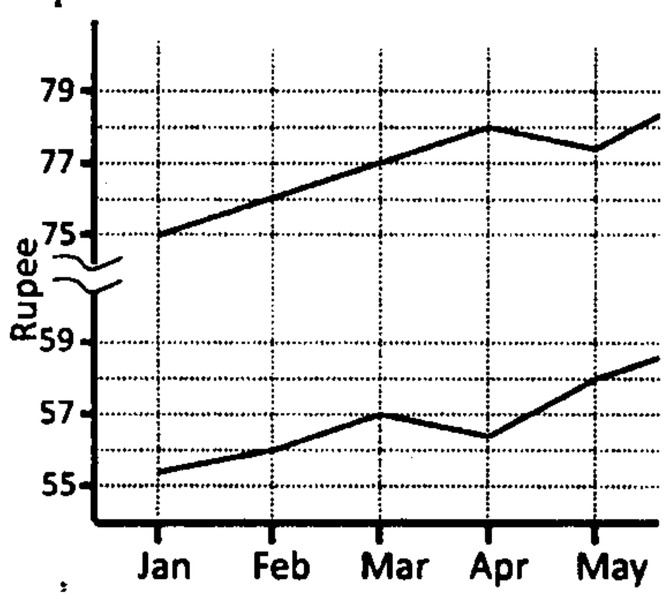
1. 1022121
2. 2042122
3. 3063126
4. 4083128

19. What is the 94th term of the following sequence?

1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, ...

1. 8
2. 9
3. 10
4. 11

20. The following graphs depict variation in the value of Dollar and Euro in terms of the Rupee over six months.



Which of the following statements is true?

1. Values of Dollar and Euro rose steadily from January to June
2. Values of Dollar and Euro rose by equal rate between January to March
3. The rise in the value of Dollar from April to May is three times the fall in Euro during the same period
4. Values of Dollar and Euro rose equally between May and June

PART 'B'

Unit- 1

21. Let $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers. Then a necessary and sufficient condition for the sequence of polynomials $f_n(x) = b_n x + c_n x^2$ to converge uniformly to 0 on the real line is

1. $\lim_{n\to\infty} b_n = 0$ and $\lim_{n\to\infty} c_n = 0$
2. $\sum_{n=1}^{\infty} |b_n| < \infty$ and $\sum_{n=1}^{\infty} |c_n| < \infty$
3. There exists a positive integer N such that $b_n = 0$ and $c_n = 0$ for all $n > N$
4. $\lim_{n\to\infty} c_n = 0$

22. Let k be a positive integer. The radius of convergence of the series

$\sum_{n=0}^{\infty} \frac{(n!)^k}{(kn)!} z^n$ is

1. k
2. k^{-k}
3. k^k
4. ∞

23. Suppose p is a polynomial with real coefficients. Then which of the following statements is necessarily true?

1. There is no root of the derivative p' between two real roots of the polynomial p
2. There is exactly one root of the derivative p' between any two real roots of p
3. There is exactly one root of the derivative p' between any two consecutive roots of p
4. There is at least one root of the derivative p' between any two consecutive roots of p

24. Let $G = \{(x, f(x)) : 0 \leq x \leq 1\}$ be the graph of a real valued differentiable function f. Assume that $(1, 0) \in G$. Suppose that the tangent vector to G at any point is perpendicular to the radius vector at that point. Then which of the following is true?

1. G is the arc of an ellipse
2. G is the arc of a circle
3. G is a line segment
4. G is the arc of a parabola

25. Let $\Omega \subseteq \mathbb{R}^n$ be an open set and $f : \Omega \to \mathbb{R}$ be a differentiable function such that $(Df)(x) = 0$ for all $x \in \Omega$.

Then which of the following is true?

1. f must be a constant function
2. f must be constant on connected components of Ω
3. $f(x) = 0$ or 1 for $x \in \Omega$
4. The range of the function f is a subset of $\mathbb{Z}$

26. Let $\{a_n : n \geq 1\}$ be a sequence of real numbers such that $\sum_{n=1}^{\infty} a_n$ is convergent and $\sum_{n=1}^{\infty} |a_n|$ is divergent. Let R be the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$. Then we can conclude that

1. $0 < R < 1$
2. $R = 1$
3. $1 < R < \infty$
4. $R = \infty$

27. Let A, B be $n \times n$ matrices such that $BA + B^2 = I - BA^2$ where I is the $n \times n$ identity matrix. Which of the following is always true?

1. A is nonsingular
2. B is nonsingular
3. A+B is nonsingular
4. AB is nonsingular

28. Which of the following matrices has the same row space as the matrix

$$\begin{pmatrix} 4 & 8 & 4 \\ 3 & 6 & 1 \\ 2 & 4 & 0 \end{pmatrix}?$$

1. $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
2. $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
3. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
4. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

29. The determinant of the $n \times n$ permutation matrix

$$\begin{bmatrix} & & & 1 \\ & & 1 & \\ & \iddots & & \\ 1 & & & \end{bmatrix}$$

1. $(-1)^n$
2. $(-1)^{\left\lfloor \frac{n}{2} \right\rfloor}$
3. -1
4. 1

Here $\lfloor x \rfloor$ denotes the greatest integer not exceeding x.

30. The determinant

$$\begin{vmatrix} 1 & 1+x & 1+x+x^2 \\ 1 & 1+y & 1+y+y^2 \\ 1 & 1+z & 1+z+z^2 \end{vmatrix}$$

is equal to

1. $(z-y)(z-x)(y-x)$
2. $(x-y)(x-z)(y-z)$
3. $(x-y)^2(y-z)^2(z-x)^2$
4. $(x^2-y^2)(y^2-z^2)(z^2-x^2)$

31. Which of the following matrices is not diagonalizable over $\mathbb{R}$?

1. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
2. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$
3. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
4. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

32. Let P be a 2×2 complex matrix such that P^*P is the identity matrix, where P^* is the conjugate transpose of P. Then the eigenvalues of P are

1. real
2. complex conjugates of each other
3. reciprocals of each other
4. of modulus 1

Unit- 2

33. Let $p(z) = a_0 + a_1 z + \cdots + a_n z^n$ and $q(z) = b_1 z + b_2 z^2 + \cdots + b_n z^n$ be complex polynomials. If a_0, b_1 are non-zero complex numbers then the residue of $\frac{p(z)}{q(z)}$ at 0 is equal to

1. $\frac{a_0}{b_1}$ 2. $\frac{b_1}{a_0}$
3. $\frac{a_1}{b_1}$ 4. $\frac{a_0}{a_1}$

34. Let $\sum_0^\infty a_n z^n$ be a convergent power series such that $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = R > 0$.
Let p be a polynomial of degree d. Then the radius of convergence of the power series $\sum_{n=0}^\infty p(n) a_n z^n$ equals
1. R 2. d
3. Rd 4. R+d

35. Let $\mathbb{Z}_+ = \{1,2, \dots\}$ be the set of positive integers. Let $\tau_1 :=$ subspace topology on $\mathbb{Z}_+$ induced from the usual topology of $\mathbb{R}$, $\tau_2 :=$ order topology on $\mathbb{Z}_+$, i.e., the topology with base
$\{\{x: 1 \le x < b\}: b \in \mathbb{Z}_+\} \cup$
$\{\{x: a < x < b\}: a, b \in \mathbb{Z}_+\}$,
$\tau_3 :=$ discrete topology.
Then
1. $\tau_1 \neq \tau_3$ and $\tau_1 = \tau_2$
2. $\tau_1 \neq \tau_2$ and $\tau_1 = \tau_3$
3. $\tau_1 \neq \tau_3$ and $\tau_2 = \tau_3$
4. $\tau_1 = \tau_2 = \tau_3$

36. The number of conjugacy classes in the permutation group S_6 is:
1. 12 2. 11
3. 10 4. 6

37. Find the degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt[4]{2}, \sqrt[8]{2})$ over $\mathbb{Q}$.
1. 4 2. 8
3. 14 4. 32

38. Let G be the Galois group of a field with nine elements over its subfield with three elements. Then the number of orbits for the action of G on the field with nine elements is
1. 3 2. 5
3. 6 4. 9

39. The number of surjective maps from a set of 4 elements to a set of 3 elements is
1. 36 2. 64
3. 69 4. 81

40. In the group of all invertible 4×4 matrices with entries in the field of 3 elements, any 3-Sylow subgroup has cardinality:
1. 3 2. 81
3. 243 4. 729

Unit- 3

41. Let $P: \mathbb{R} \to \mathbb{R}$ be a polynomial of the form $P(x) = a_0 + a_1 x + a_2 x^2$, with $a_0, a_1, a_2 \in \mathbb{R}$ and $a_2 \neq 0$.
Let $E_1 = \int_0^1 P(x)dx - \frac{1}{2}(P(0) + P(1))$
$$E_2 = \int_0^1 P(x)dx - P\left(\frac{1}{2}\right)$$
If $|x|$ is the absolute value of $x \in \mathbb{R}$, then
1. $|E_1| > |E_2|$ 2. $|E_2| > |E_1|$
3. $|E_2| = |E_1|$ 4. $|E_2| = 2|E_1|$

42. For $\lambda \in \mathbb{R}$, consider the boundary value problem
$$\left.\begin{aligned} x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} + \lambda y = 0, \quad x \in [1,2] \\ y(1) = y(2) = 0 \end{aligned}\right\} - (P_\lambda)$$
Which of the following statement is true?
1. There exists a $\lambda_0 \in \mathbb{R}$ such that (P_λ) has a nontrivial solution for any $\lambda > \lambda_0$.
2. $\{\lambda \in \mathbb{R}: (P_\lambda)$ has a nontrival solution$\}$ is a dense subset of $\mathbb{R}$.
3. For any continuous function $f: [1,2] \to \mathbb{R}$ with $f(x) \neq 0$ for some $x \in [1,2]$, there exists a solution u of (P_λ) for some $\lambda \in \mathbb{R}$ such that $\int_1^2 fu \neq 0$.
4. There exists a $\lambda \in \mathbb{R}$ such that (P_λ) has two linearly independent solutions.

43. Let $y: [0, \infty) \to \mathbb{R}$ be twice continuously differentiable and satisfy

$$y(x) + \int_0^x (x - s)y(s)ds = x^3/6.$$

Then

1. $y(x) = \frac{1}{6}\int_0^x s^3 sin(x - s)ds$
2. $y(x) = \frac{1}{6}\int_0^x s^3 cos(x - s)ds$
3. $y(x) = \int_0^x s\ sin(x - s)ds$
4. $y(x) = \int_0^x s\ \ cos(x - s)ds$

44. Consider the functional

$$J(y) = y^2(1) + \int_0^1 y'^2(x)dx,$$

$$y(0) = 1$$

where $y \in C^2([0, 1])$. If y extremizes J

then

1. $y(x) = 1 - \frac{1}{2}x^2$
2. $y(x) = 1 - \frac{1}{2}x$
3. $y(x) = 1 + \frac{1}{2}x$
4. $y(x) = 1 + \frac{1}{2}x^2$

45. Let $u(x, t) = e^{i\omega x}v(t)$ with $v(0) = 1$ be a solution to

$$\frac{\partial u}{\partial t} = \frac{\partial^3 u}{\partial x^3}$$

Then

1. $u(x, t) = e^{i\omega(x-\omega^2 t)}$
2. $u(x, t) = e^{i\omega x-\omega^2 t}$
3. $u(x, t) = e^{i\omega(x+\omega^2 t)}$
4. $u(x, t) = e^{i\omega^3(x-t)}$

46. Let $y: \mathbb{R} \to \mathbb{R}$ be differentiable and satisfy the ODE:

$$\left.\begin{aligned}\frac{dy}{dx} = f(y),\ x \in \mathbb{R}\\ y(0) = y(1) = 0\end{aligned}\right\}$$

where $f: \mathbb{R} \to \mathbb{R}$ is a Lipschitz continuous function.

Then

1. $y(x) = 0$ if and only if $x \in \{0, 1\}$
2. y is bounded
3. y is strictly increasing
4. $\frac{dy}{dx}$ is unbounded

47. The Charpit's equations for the PDE

$$up^2 + q^2 + x + y = 0,\ \ p = \frac{\partial u}{\partial x}, q = \frac{\partial u}{\partial y}$$

are given by

1. $\frac{dx}{-1-p^3} = \frac{dy}{-1-qp^2} = \frac{du}{2p^2u+2q^2} = \frac{dp}{2pu} = \frac{dq}{2q}$
2. $\frac{dx}{2pu} = \frac{dy}{2q} = \frac{du}{2p^2u+2q^2} = \frac{dp}{-1-p^3} = \frac{dq}{-1-qp^2}$
3. $\frac{dx}{up^2} = \frac{dy}{q^2} = \frac{du}{0} = \frac{dp}{x} = \frac{dq}{y}$
4. $\frac{dx}{2q} = \frac{dy}{2pu} = \frac{du}{x+y} = \frac{dp}{p^2} = \frac{dq}{qp^2}$

48. Consider a body of unit mass falling freely from rest under gravity with velocity v. If the air resistance retards the acceleration by cv where c is a constant, then

1. $v = \frac{g}{c}[1 + e^{ct}]$
2. $v = \frac{g}{c}[1 + e^{-ct}]$
3. $v = \frac{g}{c}[1 - e^{-ct}]$
4. $v = \frac{g}{c}[1 - e^{ct}]$

Unit- 4

49. Let $X_1, X_2, \ldots$ be independent and identically distributed random variables with $\mathrm{E}(X_i) = 0$ and $\mathrm{Var}(X_i) = 1$ for all i. Let $S_n = X_1 + \cdots + X_n$. Let $\Phi(x)$ denote the cumulative distribution function of a standard normal random variable. Then, for any $x > 0$, $\lim_{n\to\infty} P(-nx < S_n < nx)$ equals

1. $2\Phi(x) - 1$
2. $\Phi(x)$
3. 1
4. $1 - \Phi(2x)$

50. Five persons A, B, C, D and E are seated at random on eight numbered chairs which are arranged in a circle. What is the probability that A and B are separated by at least 2 chairs?

1. 3/7
2. 1/2
3. 4/7
4. 1/4

51. Let X and Y be integer-valued, bounded random variables. Then which of the following statement is FALSE?

1. $E(X) = \sum_y E(X|Y = y)P(Y = y)$
2. $V(X) = \sum_y V(X|Y = y)P(Y = y)$
3. $P(X = x) = \sum_y P(X = x|Y = y)P(Y = y)$
4. $E(XY) = \sum_y yE(X|Y = y)P(Y = y)$

52. Suppose $X_1, X_2, \ldots$ are random variables on a common probability space with $X_n \sim N(\mu_n, \sigma_n^2)$. Then, X_n converges in probability to 2 if and only if

1. $\mu_n \to 0$ and $\sigma_n^2 \to 2$
2. $\mu_n \to 2$ and $\sigma_n^2 \to 0$
3. $\mu_n \to 0$ and σ_n^2 converges
4. $\sigma_n^2 \to 0$ and μ_n converges

53. Suppose that X_1, X_2 and X_3 are independent and identically distributed random variables, each having a Bernoulli distribution with parameter 1/2. Consider the 2×2 matrix $A = \begin{pmatrix} X_1 & 0 \\ X_2 & X_3 \end{pmatrix}$. Then, ($A$ is invertible) equals

1. 0
2. 1
3. 1/4
4. 3/4

54. Suppose $X_1, X_2, \ldots X_n$ is a random sample from $U(0, \theta), \theta > 0$. Let $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ be the order statistics. Consider the two unbiased estimators for θ: $T_1 = 2\bar{X}$ and $T_2 = \left(\frac{n+1}{n}\right) X_{(n)}$. Then $\lim_{n\to\infty} \frac{\mathrm{Var}(T_2)}{\mathrm{Var}(T_1)} =$

1. 0
2. 1
3. ∞
4. 12

55. Suppose $X \sim$ Poisson $(\lambda), \lambda > 0$. Let the prior distribution of λ be $U(0, 1)$. If $X = 0$ is observed, then the posterior probability of the set $0 < \lambda \le \frac{1}{2}$ is

1. 0.5
2. 1
3. $\frac{e-\sqrt{e}}{e-1}$
4. $\frac{1}{e}$

56. Suppose $X \sim N(0,1)$ and the conditional distribution of Y given $X = x$ is $N(\alpha x, 1)$, for $0 < \alpha < 1$. When we regress Y on X, the coefficient of determination R^2 is

1. α^2
2. α
3. $\frac{\alpha}{\sqrt{1+\alpha^2}}$
4. $\frac{\alpha^2}{1+\alpha^2}$

57. Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be a random sample of size n from a p-variate Normal distribution with mean $\boldsymbol{\mu}$ and positive definite covariance matrix $\boldsymbol{\Sigma}$. Choose the correct statement

1. $(\boldsymbol{X}_1 - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{X}_1 - \boldsymbol{\mu})$ has chi-square distribution with 1 d.f.
2. $\bar{\boldsymbol{X}}\bar{\boldsymbol{X}}'$ has Wishart distribution with p d.f.

3. $\sum_{i=1}^{n}(X_i - \mu)(X_i - \mu)'$ has Wishart distribution with n d.f.
4. $X_1 + X_2$ and $X_1 - X_2$ are independently distributed.

58. How many distinct samples of size n can be drawn with replacement from the population $\{u_1, u_2, \ldots, u_n\}$ of n units?

1. n^n
2. $\binom{2n-1}{n}$
3. $\binom{2n-1}{n+1}$
4. 1

59. For testing the effectiveness of four teaching techniques, five teachers of a college were involved. A class of 120 students was divided into 5 groups of 24 each at random; one group was assigned to each of the five teachers. Each teacher further divided his/her group into four equal subgroups at random, and used one technique per subgroup. All of them used the same course material. After all the classes were over, a common examination was conducted and the marks were noted. Suppose we want to test whether all the four teaching techniques are equally effective. What is the degrees of freedom associated with the residual sum of squares?

1. 60
2. 100
3. 119
4. 460

60. Consider the linear programming problem:
Minimize: $z = -2x - 5y$
subject to $3x + 4y \geq 5, x \geq 0, y \geq 0$.
which of the following is correct?

1. Set of feasible solutions is empty.
2. Set of feasible solutions is non-empty but there is no optimal solution.
3. Optimal value is attained at $\left(0, \frac{5}{4}\right)$.
4. Optimal value is attained at $\left(\frac{5}{3}, 0\right)$.

PART 'C'

Unit- 1

61. Let E be a subset of $\mathbb{R}$. Then the characteristic function $\chi_E: \mathbb{R} \to \mathbb{R}$ is continuous if and only if

1. E is closed
2. E is open
3. E is both open and closed
4. E is neither open nor closed

62. Suppose that P is a monic polynomial of degree n in one variable with real coefficients and K is a real number. Then which of the following statements is/are necessarily true?

1. If n is even and $K > 0$, then there exists $x_0 \in \mathbb{R}$ such that $P(x_0) = K e^{x_0}$
2. If n is odd and $K < 0$, then there exists $x_0 \in \mathbb{R}$ such that $P(x_0) = K e^{x_0}$
3. For any natural number n and $0 < K < 1$, there exists $x_0 \in \mathbb{R}$ such that $P(x_0) = K e^{x_0}$
4. If n is odd and $K \in \mathbb{R}$, then there exists $x_0 \in \mathbb{R}$ such that $P(x_0) = K e^{x_0}$

63. Let $\{a_k\}$ be an unbounded, strictly increasing sequence of positive real numbers and $x_k = (a_{k+1} - a_k)/a_{k+1}$. Which of the following statements is/are correct?

1. For all $n \geq m$, $\sum_{k=m}^{n} x_k > 1 - \frac{a_m}{a_n}$
2. There exists $n \geq m$ such that $\sum_{k=m}^{n} x_k > \frac{1}{2}$
3. $\sum_{k=1}^{\infty} x_k$ converges to a finite limit.
4. $\sum_{k=1}^{\infty} x_k$ diverges to ∞

64. For a non-empty subset S and a point x in a connected metric space (X, d), let $d(x, S) = \inf\{d(x, y): y \in S\}$.

Which of the following statements is/are correct?

1. If S is closed and $d(x,S) > 0$ then x is not an accumulation point of S
2. If S is open and $d(x,S) > 0$ then x is not an accumulation point of S
3. If S is closed and $d(x,S) > 0$ then S does not contain x
4. If S is open and $d(x,S) = 0$ then $x \in S$

65. Let f be a continuously differentiable function on $\mathbb{R}$. Suppose that

$$L = \lim_{x\to\infty} (f(x) + f'(x))$$

exists. If $0 < L < \infty$, then which of the following statements is/are correct?

1. If $\lim_{x\to\infty} f'(x)$ exists, then it is 0
2. If $\lim_{x\to\infty} f(x)$ exists then it is L
3. If $\lim_{x\to\infty} f'(x)$ exists then $\lim_{x\to\infty} f(x) = 0$
4. If $\lim_{x\to\infty} f(x)$ exists then $\lim_{x\to\infty} f'(x) = L$

66. Let A be a subset of $\mathbb{R}$. Which of the following properties imply that A is compact?

1. Every continuous function f from A to $\mathbb{R}$ is bounded
2. Every sequence $\{x_n\}$ in A has a convergent subsequence converging to a point in A
3. There exists a continuous function from A onto $[0,1]$
4. There is no one-one and continuous function from A onto $(0,1)$

67. Let f be a monotonically increasing function from $[0,1]$ into $[0,1]$. Which of the following statements is/are true?

1. f must be continuous at all but finitely many points in $[0,1]$
2. f must be continuous at all but countably many points in $[0,1]$
3. f must be Riemann integrable
4. f must be Lebesgue integrable

68. Consider the normed linear spaces $X_1 = (C[0,1], \|\cdot\|_1)$ and $X_\infty = (C[0,1], \|\cdot\|_\infty)$, where $C[0,1]$ denotes the vector space of all continuous real valued functions on $[0,1]$ and $\|f\|_1 = \int_0^1 |f(t)|dt$, $\|f\|_\infty = \sup\{|f(t)| \mid t \in [0,1]\}$. Let U_1 and U_∞ be the open unit balls in X_1 and X_∞ respectively. Then

1. U_∞ is a subset of U_1
2. U_1 is a subset of U_∞
3. U_∞ is equal to U_1
4. Neither U_∞ is a subset of U_1 nor U_1 is a subset of U_∞

69. Let X be a metric space and $f: X \to \mathbb{R}$ be a continuous function. Let $G = \{(x, f(x): x \in X\}$ be the graph of f. Then

1. G is homeomorphic to X
2. G is homeomorphic to $\mathbb{R}$
3. G is homeomorphic to $X \times \mathbb{R}$
4. G is homeomorphic to $\mathbb{R} \times X$

70. Let A be a real $n \times n$ orthogonal matrix, that is, $A^tA = AA^t = I_n$, the $n \times n$ identity matrix. Which of the following statements are necessarily true?

1. $\langle Ax, Ay\rangle = \langle x, y\rangle \forall x, y \in \mathbb{R}^n$
2. All eigenvalues of A are either $+1$ or -1
3. The rows of A form an orthonormal basis of $\mathbb{R}^n$
4. A is diagonalizable over $\mathbb{R}$

71. Which of the following matrices have Jordan canonical form equal to

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}?$$

1. $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
2. $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
3. $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
4. $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

72. Let A be a 3×4 and b be a 3×1 matrix with integer entries. Suppose that the system $Ax = b$ has a complex solution. Then

1. $Ax = b$ has an integer solution.
2. $Ax = b$ has a rational solution.
3. The set of real solutions to $Ax = 0$ has a basis consisting of rational solutions.
4. If $b \neq 0$ then A has positive rank.

73. Let f be a non-zero symmetric bilinear form on $\mathbb{R}^3$. Suppose that there exist linear transformations $T_i: \mathbb{R}^3 \to \mathbb{R}, i = 1,2$ such that for all $\alpha, \beta \in \mathbb{R}^3$, $f(\alpha, \beta) = T_1(\alpha)T_2(\beta)$. Then:

1. rank $f = 1$
2. dim $\{\beta \in \mathbb{R}^3: f(\alpha, \beta) = 0$ for all $\alpha \in \mathbb{R}^3\} = 2$
3. f is a positive semi-definite or negative semi-definite
4. $\{\alpha: f(\alpha, \alpha) = 0\}$ is a linear subspace of dimension 2

74. The matrix $A = \begin{pmatrix} 5 & 9 & 8 \\ 1 & 8 & 2 \\ 9 & 1 & 0 \end{pmatrix}$ satisfies:

1. A is invertible and the inverse has all integer entries.
2. det (A) is odd.
3. det (A) is divisible by 13.
4. det (A) has at least two prime divisors.

75. Let A be 5×5 matrix and let B be obtained by changing one element of A. Let r and s be the ranks of A and B respectively. Which of the following statements is/are correct?

1. $s \leq r + 1$
2. $r-1 \leq s$
3. $s = r - 1$
4. $s \neq r$

76. Let $M_n(K)$ denote the space of all $n \times n$ matrices with entries in a field K. Fix a non-singular matrix $A = (A_{ij}) \in M_n(K)$, and consider the linear map $T: M_n(K) \to M_n(K)$ given by :

$$T(X) = AX.$$

Then:

1. trace $(T) = n \sum_{i=1}^{n} A_{ii}$
2. trace $(T) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij}$
3. rank of T is n^2
4. T is non-singular

77. For arbitrary subspaces U, V and W of a finite dimensional vector space, which of the following hold:

1. $U \cap (V + W) \subset U \cap V + U \cap W$
2. $U \cap (V + W) \supset U \cap V + U \cap W$
3. $(U \cap V) + W \subset (U + W) \cap (V + W)$
4. $(U \cap V) + W \supset (U + W) \cap (V + W)$

78. Let A be a 4×7 real matrix and B be a 7×4 real matrix such that $AB = I_4$, where I_4 is the 4×4 identity matrix. Which of the following is/are always true?

1. rank $(A) = 4$
2. rank $(B) = 7$
3. nullity $(B) = 0$
4. $BA = I_7$, where I_7 is the 7×7 identity matrix

Unit - 2

79. Let f be an entire function on $\mathbb{C}$ and let Ω be a bounded open subset of $\mathbb{C}$.
Let $S = \{Re\, f(z) + Im\, f(z) \mid z \in \Omega\}$.
Which of the following statements is/are necessarily correct?

1. S is an open set in $\mathbb{R}$
2. S is a closed set in $\mathbb{R}$
3. S is an open set of $\mathbb{C}$
4. S is a discrete set in $\mathbb{R}$

80. Let $u(x + iy) = x^3 - 3xy^2 + 2x$. For which of the following functions v, is $u + iv$ a holomorphic function on $\mathbb{C}$?

1. $v(x+iy) = y^3 - 3x^2y + 2y$
2. $v(x+iy) = 3x^2y - y^3 + 2y$
3. $v(x+iy) = x^3 - 3xy^2 + 2x$
4. $v(x+iy) = 0$

81. Let f be an entire function on $\mathbb{C}$. Let $g(z) = \overline{f(\bar{z})}$. Which of the following statements is/are correct?

1. if $f(z) \in \mathbb{R}$ for all $z \in \mathbb{R}$ then
   $f = g$
2. if $f(z) \in \mathbb{R}$ for all
   $z \in \{z|Im\, z = 0\} \cup \{z|Im\, z = a\}$,
   for some $a > 0$, then
   $f(z+ia) = f(z-ia)$ for all
   $z \in \mathbb{C}$
3. if $f(z) \in \mathbb{R}$ for all
   $z \in \{z|Im\, z = 0\} \cup \{z|Im\, z = a\}$,
   for some $a > 0$, then
   $f(z+2ia) = f(z)$ for all $z \in \mathbb{C}$.
4. if $f(z) \in \mathbb{R}$ for all
   $z \in \{z|Im\, z = 0\} \cup \{z|Im\, z = a\}$
   for some $a > 0$, then
   $f(z+ia) = f(z)$ all $z \in \mathbb{C}$.

82. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be an entire function and let r be a positive real number. Then

1. $\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq \sup_{|z|=r} |f(z)|^2$
2. $\sup_{|z|=r} |f(z)|^2 \leq \sum_{n=0}^{\infty} |a_n|^2 r^{2n}$
3. $\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq \frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^2 d\theta$
4. $\sup_{|z|=r} |f(z)|^2 \leq \frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^2 d\theta$

83. Let $X = \{(a,b) \in \mathbb{R}^2 : a^2 + b^2 = 1\}$ be the unit circle inside $\mathbb{R}^2$. Let $f: X \to \mathbb{R}$ be a continuous function. Then:

1. Image (f) is connected.
2. Image (f) is compact.
3. The given information is not sufficient to determine whether image (f) is bounded.
4. f is not injective.

84. Let $\mathbb{R}[x]$ denote the vector space of all real polynomials. Let $D: \mathbb{R}[x] \to \mathbb{R}[x]$ denote the map $Df = \frac{df}{dx}, \forall f$. Then,

1. D is one-one.
2. D is onto.
3. There exists $E: \mathbb{R}[x] \to \mathbb{R}[x]$ so that $D(E(f)) = f, \forall f$.
4. There exists $E: \mathbb{R}[x] \to \mathbb{R}[x]$ so that $E(D(f)) = f, \forall f$.

85. Let G be a nonabelian group. Then, its order can be:

1. 25
2. 55
3. 125
4. 35

86. Which of the following are eigenvalues of the matrix

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}?$$

1. $+1$
2. -1
3. $+i$
4. $-i$

87. Let $\mathbb{R}[x]$ be the polynomial ring over $\mathbb{R}$ in one variable. Let $I \subseteq \mathbb{R}[x]$ be an ideal. Then

1. I is a maximal ideal if and only if I is a non-zero prime ideal
2. I is a maximal ideal if and only if the quotient ring $\mathbb{R}[x]/I$ is isomorphic to $\mathbb{R}$

3. I is a maximal ideal if and only if $I = (f(x))$, where $f(x)$ is a non-constant irreducible polynomial over $\mathbb{R}$
4. I is a maximal ideal if and only if there exists a nonconstant polynomial $f(x) \in I$ of degree ≤ 2

88. Let G be a group of order 45. Then
1. G has an element of order 9
2. G has a subgroup of order 9
3. G has a normal subgroup of order 9
4. G has a normal subgroup of order 5

89. Which of the following is/are true?
1. Given any positive integer n, there exists a field extension of $\mathbb{Q}$ of degree n.
2. Given a positive integer n, there exist fields F and K such that $F \subseteq K$ and K is Galois over F with $[K:F] = n$.
3. Let K be a Galois extension of $\mathbb{Q}$ with $[K:\mathbb{Q}] = 4$. Then there is a field L such that $K \supseteq L \supseteq \mathbb{Q}, [L:\mathbb{Q}] = 2$ and L is a Galois extension of $\mathbb{Q}$.
4. There is an algebraic extension K of $\mathbb{Q}$ such that $[K:\mathbb{Q}]$ is not finite.

90. Let $A = \begin{pmatrix} x & y \\ -y & x \end{pmatrix}$, where $x, y \in \mathbb{R}$ such that $x^2 + y^2 = 1$. Then we must have:
1. For any $n \geq 1$, $A^n = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ where $x = \cos(\theta/n), y = \sin(\theta/n)$
2. $tr(A) \neq 0$
3. $A^t = A^{-1}$
4. A is similar to a diagonal matrix over $\mathbb{C}$

Unit - 3

91. The system of ODE
$$\left.\begin{aligned} \frac{dx}{dt} &= (1+x^2)y,\ t \in \mathbb{R} \\ \frac{dy}{dt} &= -(1+x^2)x,\ t \in \mathbb{R} \\ (x(0), y(0)) &= (a, b) \end{aligned}\right\}$$
has a solution:
1. only if $(a, b) = (0,0)$
2. for any $(a, b) \in \mathbb{R} \times \mathbb{R}$
3. such that $x^2(t) + y^2(t) = a^2 + b^2$ for all $t \in \mathbb{R}$
4. such that $x^2(t) + y^2(t) \to \infty$ as $t \to \infty$ if $a > 0$ and $b > 0$

92. Let $y: \mathbb{R} \to \mathbb{R}$ be a solution of the ODE
$$\left.\begin{aligned} \frac{d^2y}{dx^2} - y &= e^{-x}, x \in \mathbb{R} \\ y(0) = \frac{dy}{dx}(0) &= 0 \end{aligned}\right\} \text{ then}$$
1. y attains its minimum on $\mathbb{R}$
2. y is bounded on $\mathbb{R}$
3. $\lim_{x\to\infty} e^{-x}y(x) = \frac{1}{4}$
4. $\lim_{x\to-\infty} e^{x}y(x) = \frac{1}{4}$

93. Let $u \in c^2([0,1])$ satisfy for some $\lambda \neq 0$ and $a \neq 0$
$u(x) + \frac{\lambda}{2}\int_0^1 |x-s|u(s)ds = ax + b$
Then u also satisfies
1. $\frac{d^2u}{dx^2} + \lambda u = 0$
2. $\frac{d^2u}{dx^2} - \lambda u = 0$
3. $\frac{du}{dx} - \frac{\lambda}{2}\int_0^1 \frac{x-s}{|x-s|}u(s)ds = a$
4. $\frac{du}{dx} + \frac{\lambda}{2}\int_0^1 \frac{x-s}{|x-s|}u(s)ds = a$

94. Let $f: \mathbb{R} \to \mathbb{R}$ be a smooth function with non-vanishing derivative. The Newton's method for finding a root of $f(x) = 0$ is the same as

1. is necessarily bounded
2. satisfies $|u(x,t)| < e^t$
3. is necessarily unbounded
4. is oscillatory in x.

101. Let $u = u(x,t)$ be the solution of the Cauchy problem

$$\frac{\partial u}{\partial t} + \left(\frac{\partial u}{\partial x}\right)^2 = 1 \qquad x \in \mathbb{R}, t > 0$$

$$u(x,0) = -x^2 \qquad x \in \mathbb{R}$$

Then

1. $u(x,t)$ exists for all $x \in \mathbb{R}$ and $t > 0$.
2. $|u(x,t)| \to \infty$ as $t \to t^*$ for some $t^* > 0$ and $x \neq 0$
3. $u(x,t) \leq 0$ for all $x \in \mathbb{R}$ and for all $t < 1/4$.
4. $u(x,t) > 0$ for some $x \in \mathbb{R}$ and $0 < t < 1/4$.

102. Let $y(t)$ satisfy the differential equation $y' = \lambda y; y(0) = 1$.
Then the backward Euler method, for $n \geq 1$ and $h > 0$
$\frac{y_n - y_{n-1}}{h} = \lambda y_n;\ y_0 = 1$ yields

1. a first order approximation to $e^{\lambda nh}$
2. a polynomial approximation to $e^{\lambda nh}$
3. a rational function approximation to $e^{\lambda nh}$
4. a Chebyshev polynomial approximation to $e^{\lambda nh}$

Unit - 4

103. Suppose X is a random variable such that $E(X) = 0, E(X^2) = 2$ and $E(X^4) = 4$. Then

1. $E(X^3) = 0$
2. $P(X \geq 0) = 1/2$
3. $X \sim N(0,2)$
4. X is bounded with probability 1.

104. Suppose X_1, X_2, X_3 and X_4 are independent and identically distributed random variables, having density function f. Then,

1. $P(X_4 > \max(X_1, X_2) > X_3) = \frac{1}{6}$
2. $P(X_4 > \max(X_1, X_2) > X_3) = \frac{1}{8}$
3. $P(X_4 > X_3 > \max(X_1, X_2)) = \frac{1}{12}$
4. $P(X_4 > X_3 > \max(X_1, X_2)) = \frac{1}{6}$

105. Let $X_1, X_2, X_3, \ldots$ be independent random variables with $E(X_k) = 0$ and $\mathrm{Var}(X_k) = k$.
Let $S_n = \sum_{k=1}^{n} X_k$ .Then, as $n \to \infty$,

1. $\frac{S_n}{n^{3/2}} \to 0$ in probability
2. $\frac{S_n}{n^{3/2}} \to 0$ in distribution
3. $\frac{S_n X_n}{n^{5/2}} \to 0$ in distribution
4. $\frac{S_n X_n}{n^{5/2}} \to 0$ in probability

106. Consider a Markov chain with state space $\{1, 2, \ldots, 100\}$. Suppose states $2i$ and $2j$ communicate with each other and states $2i - 1$ and $2j - 1$ communicate with each other for every $i, j = 1, 2, \ldots, 50$. Further suppose that $p_{3,3}^{(2)} > 0, p_{4,4}^{(3)} > 0$ and $p_{2,5}^{(7)} > 0$. Then

1. The Markov chain is irreducible.
2. The Markov chain is aperiodic.
3. State 8 is recurrent.
4. State 9 is recurrent.

107. For a Markov chain with finite state space, the number of stationary distributions can be

1. 0
2. 1
3. 2
4. ∞

108. $N, A_1, A_2, \ldots$ are independent real valued random variables such that

$P(N = k) = (1-p)p^k, k = 0,1,2, \ldots$

where $0 < p < 1$, and $\{A_i : i = 1,2, \ldots\}$ is a sequence of independent and identically distributed bounded random variables. Let

$$X(w) = \begin{cases} 0 & \text{if } N(w) = 0 \\ \sum_{j=1}^{k} A_j & \text{if } N(w) = k, k = 1,2, \ldots \end{cases}$$

Which of the following are necessarily correct?

1. X is a bounded random variable
2. Moment generating function m_X of X is
$$m_X(t) = \frac{(1-p)}{1-pm_A(t)}, t \in \mathbb{R},$$
where m_A is the moment generating function of A_1.
3. Characteristic function φ_X of X is $\varphi_X(t) = \frac{(1-p)}{1-p\varphi_A(t)}, t \in \mathbb{R}$, where φ_A is the characteristic function of A_1.
4. X is symmetric about 0.

109. Let $\phi(t)$ be a characteristic function of some random variable. Then, which of the following are also characteristic function?

1. $f(t) = [\phi(t)]^2$ for all $t \in \mathbb{R}$.
2. $f(t) = |\phi(t)|^2$ for all $t \in \mathbb{R}$.
3. $f(t) = \phi(-t)$ for all $t \in \mathbb{R}$.
4. $f(t) = \phi(t+1)$ for all $t \in \mathbb{R}$.

110. Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed observations from the distribution with density

$$f(x|\mu) = e^{-(x-\mu)}, x > \mu$$

where $-\infty < \mu < \infty$ and μ is unknown. Let $T_1 = 2\sum_{i=1}^{n} X_i$ and $T_2 = 2X_{(1)}$ where $X_{(1)}$ is the smallest order statistic. To test $H_0: \mu = 0$ versus $H_1: \mu > 0$ at level α, where $0 < \alpha < 1$, consider the two tests A and B given below.

A: Reject H_0 if $T_1 > C_1$ where C_1 is such that $P(Y_1 > C_1) = \alpha$ with $Y_1 \sim \chi^2_{2n}$

B: Reject H_0 if $T_2 > C_2$ where C_2 is such that $P(Y_2 > C_2) = \alpha$ with $Y_2 \sim \chi^2_2$

Then which of the following statements are valid?

1. Both A and B are level α tests
2. A is the uniformly most powerful level α test
3. B is the uniformly most powerful level α test
4. B is more powerful than A at any $\mu > 0$

111. Let $X_1, X_2, \ldots, X_n, n \geq 3$, be a random sample from $N(\mu, 1)$ population where μ is unknown. Define $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Which of the following are necessaily true?

1. $\text{Cov}(X_1 - 3X_2 + 2X_3, \bar{X}_n) = 0$
2. Cramer-Rao lower bound for unbiased estimators of μ is $\frac{1}{n}$
3. $\text{Var}(\bar{X}_n) < \text{Var}\left(\frac{X_1+2X_2+3X_3+\cdots+nX_n}{\frac{n(n+1)}{2}}\right)$
4. $\bar{X}_n$ is a function of any sufficient statistic for μ

112. A and B are two methods to determine the levels of mercury in fish. In a study to compare A and B, amount of mercury was measured using both methods on $n = 12$ fish. Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be those measurements, with X_i's standing for method A and Y_i's for method B. It should be noted that the size of error in measurement can depend on the amount of mercury, so the observations

$(X_1, Y_1), \ldots, (X_n, Y_n)$ may not be identically distributed. To test
H_0: There is no difference between methods A and B
versus
H_1: Method B typically gives a larger reading than method A,
which of the following test statistics are appropriate?

1. Number of pairs (X_i, Y_j) with $(Y_j > X_i),\ 1 \leq i \leq n; 1 \leq j \leq n$
2. Sum of the ranks of the Y observations in the combined sample
3. Number of the pairs (X_i, Y_i) with $(Y_i > X_i),\ 1 \leq i \leq n$
4. $\bar{Y} - \bar{X}$

113. Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed Bernoulli (θ), where $0 < \theta < 1$ and $n > 1$. Let the prior density of θ be proportional to $\frac{1}{\sqrt{\theta(1-\theta)}}, 0 < \theta < 1$. Define

$$S = \sum_{i=1}^{n} X_i.$$

Then valid statements among the following are:

1. The posterior mean of θ does not exist;
2. The posterior mean of θ exists;
3. The posterior mean of θ exists and it is larger than the maximum likelihood estimator for all values of S;
4. The posterior mean of θ exists and it is larger than the maximum likelihood estimator for some values of S.

114. Consider the linear model $E(\boldsymbol{Y}) = \boldsymbol{X\beta}, \mathrm{Cov}(\boldsymbol{Y}) = \sigma^2 \boldsymbol{I}$ where $\boldsymbol{X}$ is a matrix of size $n \times p$ having rank $r \leq p$. Then which of the following statements are necessarily true?

1. The set of estimable linear functions form a vector space of dimension r.
2. If $E(\boldsymbol{c'Y}) = 0$ for some nonzero vector $\boldsymbol{c}$, then there is a function $\boldsymbol{l'\beta}$ which is not estimable.
3. If all linear functions $\boldsymbol{l'\beta}$ are estimable, then $r = p$.
4. The set of functions $\boldsymbol{c'Y}$ with $E(\boldsymbol{c'Y}) = 0$ form a vector space of dimension r.

115. Consider the linear model

$$y_1 = 2\theta + \beta + \epsilon_1$$

$$y_2 = \beta + 2\gamma + \epsilon_2$$

$$y_3 = \theta + \beta + \gamma + \epsilon_3$$

where θ, β, γ are unknown parameters and $\epsilon_1, \epsilon_2, \epsilon_3$ are uncorrelated random errors with mean 0 and constant variance. Then which of the following statements are true?

1. θ, β and γ are estimable
2. $\theta - \gamma$ is estimable
3. $2\gamma - 2\theta$ is estimable
4. $\theta + \gamma$ is estimable

116. Let $\mathbf{Y}$ follow multivariate normal distribution $N_n(\mathbf{0}, \boldsymbol{I})$ and let A and B be $n \times n$ symmetric, idempotent matrices. Then which of the following statements are true?

1. If $AB = 0$, then $\boldsymbol{Y'AY}$ and $\boldsymbol{Y'BY}$ are independently distributed.
2. If $\boldsymbol{Y'}(A + B)\boldsymbol{Y}$ has chi-square distribution then $\boldsymbol{Y'AY}$ and $\boldsymbol{Y'BY}$ are independently distributed.
3. $\boldsymbol{Y'}(A - B)\boldsymbol{Y}$ has chi-square distribution.
4. $\boldsymbol{Y'AY}$ and $\boldsymbol{Y'BY}$ have chi-square distribution.

117. Consider a 3-variate population with covariance matrix $\begin{pmatrix} \sigma^2 & \sigma^2\rho & 0 \\ \sigma^2\rho & \sigma^2 & \sigma^2\rho \\ 0 & \sigma^2\rho & \sigma^2 \end{pmatrix}$ where $\sigma^2 > 0, \rho > 0$. Then which of the following statements are true?

1. $\rho < \frac{1}{\sqrt{2}}$
2. The proportion of the total population variance explained by the first principal component is $\frac{1+2\rho}{3}$
3. The second principal component is uncorrelated with the first and the third principal component.
4. The proportion of the total population variance explained by the first two principal components is $\frac{\sqrt{2}}{3}(\rho + \sqrt{2})$

118. Consider a 2^3 factorial experiment with three factors A, B and C. Suppose eight treatments are assigned in two blocks of each of the four replicates in the following way.

Replicate 1		Replicate 2		Replicate 3		Replicate 4	
(1)	b	(1)	a	(1)	a	(1)	b
a	c	b	c	c	b	ab	a
bc	ac	ac	bc	ab	ac	bc	c
abc	ab	abc	ab	abc	bc	ac	abc

Which of the following are necessarily true?

1. This is an example of complete confounding
2. AB is confounded in Replicate 1
3. AC is confounded in Replicate 2
4. ABC is confounded in Replicate 4

119. Suppose a sample of size n is drawn using simple random sampling without replacement from a finite population of N units where $N > n$ and denote the sample mean of the study variables corresponding to the selected units by $\bar{y}$. Now suppose we know one variate value y_1 corresponding to one unit and draw a simple random sample of size n without replacement from the remaining $(N-1)$ units and denote the sample mean of the study variables corresponding to the selected units by $\bar{y}_0$. Define $t_1 = N\bar{y}, t_2 = (N-1)\bar{y}_0 + y_1, V_1 = \text{Var}(t_1)$ and $V_2 = \text{Var}(t_2)$. Which of the following are necessarily true?

1. t_1 is unbiased for population total
2. t_2 is unbiased for population total
3. $V_1 = N^2 \frac{\sigma^2}{n} \frac{N-n}{N-1}$ where $\sigma^2 =$ population variance
4. $V_2 \leq V_1$ for all n, N

120. Let $X(t) =$ number of customers at time t in the system in an M/M/1 queueing model with arrival rate $\lambda > 0$ and service rate $\mu > 0$. Let $\pi_k = \lim_{t\to\infty} P(X(t) = k), k = 0, 1, 2 \cdots$ whenever it exists.

Which of the following are true?

1. $\{X(t)\}$ is a birth and death process with birth rates $\lambda_k = \lambda, k = 0, 1, 2, \ldots$ and death rates $\mu_k = \mu, k = 1, 2, \ldots$
2. $\{X(t)\}$ is a birth and death process with birth rates $\lambda_k = \frac{1}{\lambda}, k = 0, 1, 2, \ldots$ and death rates $\mu_k = \frac{1}{\mu}, k = 1, 2, \ldots$
3. Limiting distribution $\{\pi_k\}$ exists if and only if $\mu > \lambda$, and is the geometric distribution with parameter $\left(\frac{\lambda}{\mu}\right)$.
4. If an arriving customer finds exactly one customer, then his total waiting time in the system has an exponential distribution with parameter (2μ).

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